Two Wright–Fisher updates and their stationary marginals

Setup

Each individual carries a mutation count $k \in \mathbb{N}_0$. We use exponential selection $w_k = e^{-\alpha k}$ ($\alpha > 0$) and Poisson mutation at rate $\lambda \geq 0$ (per individual, per generation), so the new mutation count of an offspring is the parent's class plus an independent $\mathrm{Poi}(\lambda)$.

Let Q denote a probability distribution on $\mathbb{N}_0$ (the marginal class distribution of one individual). Two basic operators act on such Q:

$$(M_\lambda Q)(k) \;=\; (Q * \mathrm{Poi}(\lambda))(k) \;=\; \sum_{j=0}^{k} Q(j)\,\mathrm{Poi}(\lambda; k-j) \qquad \text{(mutation)},$$

$$(S\,Q)(k) \;=\; \frac{w_k\,Q(k)}{\sum_{j\geq 0} w_j\,Q(j)} \qquad \text{(selection / reweighting by } w\text{)}.$$

We compare two update rules per generation:

$$\text{(A) “sel} \to \text{mut” :} \quad T_A := M_\mu \circ S, \qquad \text{(B) “mut} \to \text{sel” :} \quad T_B := S \circ M_\lambda.$$

Multinomial resampling of N offspring is applied afterwards; for the analysis of the marginal we look at the deterministic large-N limit and return to fluctuations at the end.

Two Poisson identities

The reason the calculation is short is that both operators preserve the Poisson family:

$$S\big(\mathrm{Poi}(\nu)\big) \;=\; \mathrm{Poi}\big(\nu\, e^{-\alpha}\big), \tag{1}$$

$$M_\lambda\big(\mathrm{Poi}(\nu)\big) \;=\; \mathrm{Poi}\big(\nu + \lambda\big). \tag{2}$$

(Eq. (1) follows from $w_k\,\mathrm{Poi}(\nu; k) = e^{-\alpha k} e^{-\nu} \nu^k / k! \propto (\nu e^{-\alpha})^k / k!$, i.e. exponentially tilting Poisson rescales its parameter. Eq. (2) is the standard sum-of-independent-Poissons identity.)

Stationary Poisson marginals

Suppose the marginal is $Q = \mathrm{Poi}(\theta_{\mathrm{eq}})$ and we look for fixed points of T_A, T_B.

Model A (sel $\to$ mut). By (1) and (2),

$$T_A\big(\mathrm{Poi}(\theta_{\mathrm{eq}})\big) = M_\mu\big(\mathrm{Poi}(\theta_{\mathrm{eq}} e^{-\alpha})\big) = \mathrm{Poi}\big(\theta_{\mathrm{eq}}\, e^{-\alpha} + \mu\big).$$

The fixed-point equation $\theta_{\mathrm{eq}} = \theta_{\mathrm{eq}}\, e^{-\alpha} + \mu$ gives

$$\boxed{\theta_{\mathrm{eq}}^{(A)} \;=\; \frac{\mu}{1 - e^{-\alpha}}\,.} \tag{3}$$

Model B (mut $\to$ sel). Now

$$T_B\big(\mathrm{Poi}(\theta_{\mathrm{eq}})\big) = S\big(\mathrm{Poi}(\theta_{\mathrm{eq}} + \lambda)\big) = \mathrm{Poi}\big((\theta_{\mathrm{eq}} + \lambda)\, e^{-\alpha}\big),$$

so $\theta_{\mathrm{eq}} = (\theta_{\mathrm{eq}} + \lambda) e^{-\alpha}$ yields

$$\boxed{\theta_{\mathrm{eq}}^{(B)} \;=\; \frac{\lambda\, e^{-\alpha}}{1 - e^{-\alpha}} \;=\; \frac{\lambda}{e^{\alpha} - 1}\,.} \tag{4}$$

For small α, both formulas reduce to $\theta_{\mathrm{eq}} \approx \lambda/\alpha$, but for finite α they differ; in particular

$$\frac{\theta_{\mathrm{eq}}^{(B)}}{\theta_{\mathrm{eq}}^{(A)}} \;=\; \frac{\lambda}{\mu} \cdot \frac{1 - e^{-\alpha}}{e^{\alpha} - 1} \;=\; \frac{\lambda}{\mu} \cdot e^{-\alpha}.$$